SQL Assignment

Assignment 1

1)List all employees, ordered by their HIREDATE from newest to oldest.

SELECT EMPNO,ENAME,HIREDATE FROM EMP ORDER BY HIREDATE DESC;

2)List all employees who work in 'DALLAS'.

SELECT EMPNO,ENAME,LOC
FROM EMP JOIN DEPT
ON EMP.DEPTNO = DEPT.DEPTNO
WHERE LOC = 'DALLAS';

3)Find all employees who were hired in the year 1981.

SELECT EMPNO,ENAME,HIREDATE FROM EMP
WHERE HIREDATE >= STR_TO_DATE ('01/01/1981','%d/%m/%Y')
AND HIREDATE <= STR_TO_DATE ('31/12/1981','%d/%m/%Y');

OR

SELECT EMPNO,ENAME,HIREDATE FROM EMP
WHERE YEAR(HIREDATE) = 1981;

4)List the DNAME and LOC from the DEPT table, ordered alphabetically by location.

SELECT DNAME,LOC FROM DEPT ORDER BY LOC;

5)List all employees who do not work in Department 10.

SELECT EMPNO,ENAME,DEPTNO FROM EMP WHERE DEPTNO != 10;

6)Display all employees who were hired after January 1st, 1982.

SELECT EMPNO,ENAME,HIREDATE FROM EMP
WHERE HIREDATE > STR_TO_DATE('01/01/1982','%d/%m/%Y');

7)Show what each employee's salary would be if they were given a 15% raise. Display the ENAME, SAL, and the NEW_SAL.

SELECT ENAME,SAL,(SAL+SAL*0.15) AS NEW_SAL FROM EMP;

8)List all employees, sorted first by their Department Number (ascending) and then by their Salary (descending).

SELECT EMPNO,ENAME,DEPTNO,SAL FROM EMP
ORDER BY DEPTNO , SAL DESC;

9)List all employees whose total monthly income (Salary + Commission) is greater than 1500.

SELECT ENAME , SAL , SAL+IFNULL(COMM,0) AS TOTAL_SAL
FROM EMP WHERE (SAL+IFNULL(COMM,0))>1500;

10)Find all 'SALESMAN' who earn a salary that is not between 1200 and 1500.

SELECT ENAME, JOB, SAL FROM EMP
WHERE JOB = 'SALESMAN'
AND SAL NOT BETWEEN 1200 AND 1500;

Assignment 2

1)Find all employees whose last names start with 'A' and who
work in department No 10 or 30.

SELECT ENAME ,DEPTNO FROM EMP
WHERE ENAME LIKE 'A%'
AND (DEPTNO=10 OR DEPTNO=30);

2)Write a SQL query to display the department name, salary grade, and the total number of employees belonging to each grade within each department.

SELECT DNAME , GRADE , COUNT(*) AS TOTAL_EMP
FROM DEPT D JOIN EMP E ON D.DEPTNO = E.DEPTNO
JOIN SALGRADE S ON E.SAL BETWEEN S.LOSAL AND HISAL
GROUP BY D.DNAME , S.GRADE;

3)Retrieve the names of all employees, their department names, and their specific salary grade, but only for those employees who fall into Grade 4 or Grade 5.

SELECT ENAME , DNAME , GRADE
FROM EMP E JOIN DEPT D ON E.DEPTNO = D.DEPTNO

JOIN SALGRADE S ON E.SAL BETWEEN S.LOSAL AND S.HISAL
WHERE S.GRADE IN (4,5);

4)List the names of all employees (ENAME) along with their Department Names (DNAME), but only for those working in 'NEW YORK' or 'CHICAGO'.

SELECT ENAME , DNAME
FROM EMP E JOIN DEPT D ON E.DEPTNO = D.DEPTNO
WHERE D.LOC = 'NEW YORK' OR D.LOC = 'CHICAGO';

5)List the names of all employees who do not have a manager (their MGR column is NULL).

SELECT EMPNO , ENAME FROM EMP
WHERE MGR IS NULL;

6)Find employees whose salary is greater than 2000 and show their department names.

SELECT EMPNO , ENAME , DNAME ,SAL
FROM EMP E JOIN DEPT D ON E.DEPTNO = D.DEPTNO
WHERE E.SAL > 2000;

7)Display employees working in "ACCOUNTING" with their salary grade.

SELECT ENAME , JOB ,DNAME, GRADE FROM EMP E
JOIN SALGRADE S ON E.SAL BETWEEN S.LOSAL AND S.HISAL
JOIN DEPT D ON E.DEPTNO = D.DEPTNO
WHERE D.DNAME = 'ACCOUNTING';

8)List employee name and department name in ascending order of employee name.

SELECT ENAME , DNAME
FROM EMP E JOIN DEPT D ON
E.DEPTNO = D.DEPTNO
ORDER BY ENAME;

9)List employees working in department number 10 with their dnames and salaries and grades.

SELECT ENAME ,DNAME , SAL , GRADE
FROM EMP E JOIN DEPT D ON E.DEPTNO = D.DEPTNO
JOIN SALGRADE S ON E.SAL BETWEEN S.LOSAL AND S.HISAL
WHERE E.DEPTNO = 10;

10)Show employee name, department name, and salary sorted by salary in descending order.

SELECT ENAME,DNAME, SAL FROM EMP JOIN DEPT ON EMP.DEPTNO=DEPT.DEPTNO ORDER BY SAL DESC;

Assignment 3

1 How can you find only the departments that currently have no employees assigned to them?

SELECT D.DEPTNO, D.DNAME
FROM DEPT D
LEFT JOIN EMP E ON D.DEPTNO = E.DEPTNO
WHERE E.EMPNO IS NULL;

2.List Employees Who Are Both Managers and Analysts (Using INTERSECT)**(INTERSECT IS NOT SUPPORTED IN MYSQL)**

SELECT ENAME
FROM EMP
WHERE JOB = 'MANAGER'
AND JOB = 'ANALYST';

3.Employees Who Work in Same Department as 'SCOTT' and 'KING'(Using INTERSECT)**(INTERSECT IS NOT SUPPORTED IN MYSQL)**

SELECT ENAME
FROM EMP
WHERE DEPTNO = (SELECT DEPTNO FROM EMP WHERE ENAME='SCOTT')
AND DEPTNO = (SELECT DEPTNO FROM EMP WHERE ENAME='KING');

4.List Employees in Dept 10 or Having Salary > 3000 (Using UNION)

SELECT EMPNO , ENAME FROM EMP WHERE DEPTNO = 10
UNION
SELECT EMPNO , ENAME FROM EMP WHERE SAL > 3000;

5.Find Employees Who Do NOT Belong to Any Department

SELECT E.EMPNO, E.ENAME
FROM EMP E

LEFT JOIN DEPT D ON E.DEPTNO = D.DEPTNO
WHERE D.DEPTNO IS NULL;

6.List Departments with Total Salary Greater Than 2000

SELECT DEPTNO,SUM(SAL) AS TOTAL_SAL FROM EMP
GROUP BY DEPTNO
HAVING SUM(SAL) > 2000;

7.Show Dept with Highest Total Salary

SELECT DEPTNO, SUM(SAL) AS TOTAL_SAL
FROM EMP
GROUP BY DEPTNO
ORDER BY TOTAL_SAL DESC
LIMIT 1;

8.Display Total Commission Paid Department-wise

SELECT DEPTNO, SUM(IFNULL(COMM,0)) AS TOTAL_COMM
FROM EMP
GROUP BY DEPTNO;

9.Show Departments Where Average Salary is Greater Than Overall Average

SELECT DEPTNO , AVG(SAL) FROM EMP
GROUP BY DEPTNO
HAVING AVG(SAL) > (SELECT AVG(SAL) FROM EMP);

10.Departments Where At Least One Employee Earns More Than 4000

SELECT DEPTNO
FROM EMP
GROUP BY DEPTNO
HAVING MAX(SAL) > 4000;

Assignment 4

1) List employees who have the same job title as any employee working in 'CHICAGO'.

SELECT ENAME FROM EMP
WHERE JOB IN (SELECT JOB FROM EMP JOIN DEPT ON EMP.DEPTNO= DEPT.DEPTNO
WHERE DEPT.LOC='CHICAGO');

OR

```
SELECT DISTINCT E1.ENAME
FROM EMP E1
JOIN EMP E2 ON E1.JOB = E2.JOB
JOIN DEPT D ON E2.DEPTNO = D.DEPTNO
WHERE D.LOC = 'CHICAGO';
```

2)List the names of employees who report to a manager based in 'NEW YORK'.

```
SELECT E1.EMPNO, E1.ENAME, E1.MGR
FROM EMP E1
JOIN EMP E2 ON E1.MGR = E2.EMPNO
JOIN DEPT D ON E2.DEPTNO = D.DEPTNO
WHERE D.LOC = 'NEW YORK';
```

3)List the names and salaries of employees who earn more than the average salary of the entire company.

```
SELECT ENAME , SAL FROM EMP
WHERE SAL > (SELECT AVG(SAL) FROM EMP);
```

4)Find all employees who hold the same job as 'SMITH' but earn a different salary than him.

```
SELECT ENAME FROM EMP
WHERE JOB = (SELECT JOB FROM EMP WHERE ENAME = 'SMITH')
AND
SAL != (SELECT SAL FROM EMP WHERE ENAME = 'SMITH');
```

5)List all employees who were hired after the employee 'ADAMS' was hired.

```
SELECT ENAME FROM EMP
WHERE HIREDATE > (SELECT HIREDATE FROM EMP
WHERE ENAME = 'ADAMS');
```

6)Find the names of employees who work in the same department as the employee who earns the highest commission.

```
SELECT ENAME ,DEPTNO FROM EMP
```

```sql
WHERE DEPTNO = (
SELECT DEPTNO FROM EMP WHERE COMM = (
SELECT MAX(COMM) FROM EMP));
```

7)Show the names of employees whose salary is exactly equal to the minimum salary in the company.

```sql
SELECT ENAME FROM EMP
WHERE SAL =(
SELECT MIN(SAL) FROM EMP);
```

8)Find all employees who have the same manager as 'FORD'.

```sql
SELECT ENAME FROM EMP
WHERE MGR = (
SELECT MGR FROM EMP WHERE ENAME = 'FORD')
AND ENAME <> 'FORD';
```

9)Display the employee names who work in the same department as 'SCOTT'.

```sql
SELECT ENAME FROM EMP
WHERE DEPTNO = (
SELECT DEPTNO FROM EMP WHERE ENAME = 'SCOTT')
AND ENAME <> 'SCOTT';
```

10)Find the names of all employees hired in the same year as 'JAMES'

```sql
SELECT ENAME ,HIREDATE FROM EMP
WHERE YEAR(HIREDATE) = (
SELECT YEAR(HIREDATE) FROM EMP WHERE ENAME = 'JAMES')
AND ENAME <> 'JAMES';
```